

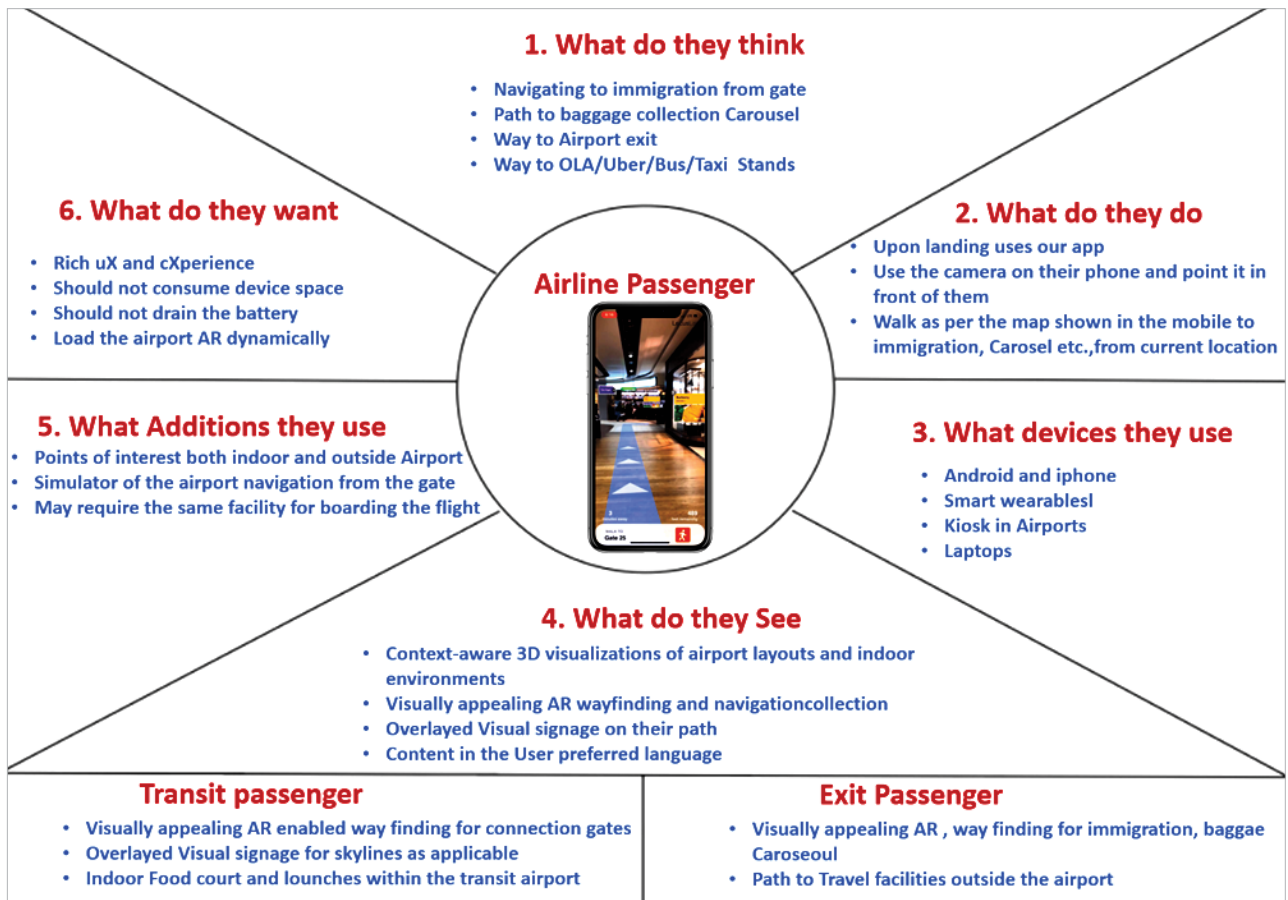


Figure 1: Overview of the system

As Shweta finally made her way out of the airport, she almost felt a sense of accomplishment at having overcome all the frustrations.

The above is one air passenger's exit experience. Even though a few hi-profile passengers use airport navigation apps, they still face difficulties such as loading different apps for each airport. So, what is the solution?

Before developing our solution let us look into some technical components considered for it. We will apply design thinking, and build a mobile app and a web app leveraging generative AI and augmented reality.

Empathy map of our solution

Our solution will provide the features shown in the figure of the empathy map (Figure 1). Our main focus will be on building the mobile app, which

will use generative AI to dynamically load the airport AR module. This module will disappear 24 hours after the flight arrives at the destination. Our mobile app will consume very little memory. We will use generative AI to provide a dynamic personalised airport navigation experience to passengers, as some passengers may need to catch different connection flights.

Mobile development approaches

We will adopt open source solutions for developing our app. Four major development approaches can be used when building mobile applications.

- Native mobile applications
- Cross-platform native mobile applications
- Hybrid mobile applications
- Progressive web applications

Each of these has its own set of advantages and disadvantages. When choosing the right development approach, we need to consider the desired user experience, the computing resources and native features required for our app, the development budget, time targets, and resources available to maintain the app.

We will opt for the cross-platform approach to create a mobile app and the web application. The latter will be used in airport kiosks and laptops to provide the airport navigation experience for passengers who don't have smartphones. We will also use augmented reality and generative AI in combination, and host the application on cloud. Our mobile solution will provide similar experiences for both Apple and Android users, and will be accessible from smart wearables also.